Lecture 1 (September 1st)

Definition. (Density operator) A general quantum state is represented by a density operator.

$$\rho = \sum_{i=1}^{n} p_i |\psi_i\rangle\langle\psi_i|$$

where ρ is

(i) Hermitian $\rho = \rho^\dagger$

(ii) Positive operator, meaning that $\langle\psi|\rho|\psi\rangle \geq 0$ for an arbitary $|\psi\rangle$

(iii) Trace is 1, that is, $\mathrm{Tr}(\rho) = 1$

It is easy to think the operator that gives classical probability weighting when a measurement is preformed. For example, take an experiment that either has a spin up or spin down as its output. Generally, an output would be expressed as a superposition,

$$\mathrm{Exp}(O) = \langle\psi_1|\psi_1\rangle p_1 \langle\psi_1|O|\psi_1\rangle + \langle\psi_2|\psi_2\rangle p_2 \langle\psi_2|O|\psi_2\rangle + \dots$$

Notice that by taking the above operator as the definition of a mixed state, we can simply take

$$\mathrm{Tr}(\rho O)$$

to obtain the expected value of a certain observable. The thing is, there are two types of uncertainty – classical and quantum.

Theorem. (Condition for pure states and equivalences $*$) Notice that a pure state can be expressed as a density operator like the following

$$\hat{\rho} = |\psi\rangle\langle\psi|$$

These types of operators, called projection operators, are idempotent

$$\rho^2 = \rho$$

meaning that their expected value is either 1 or 0. Now notice that two density matrices of pure states are equivalent up to phase. We see that

$$|x\rangle\langle x| = |y\rangle\langle y|$$

if and only if

$$|x\rangle = |y\rangle\langle y|x\rangle$$

where

$$1 = |\langle x|y\rangle|^2$$

and the states differ purely by some phase factor $\exp(i\theta)$ as we would expect.

Lecture 2 (September 3rd)

Theorem. (Born rule for density operators) Using the spectral decomposition of a certain state $A = \sum a|a\rangle\langle a| = \sum a\mathrm{Pr}_a$, we can express the probability of observing A and obtaining a as

$$\mathrm{Tr}(\rho\,\mathrm{Pr}_a)$$

Theorem. (Von Neumann equation) We check that the density matrix satisfies, for a pure state,

$$\begin{aligned}\frac{d\rho}{dt} =& \sum_i \Big(p_i|\psi_i\rangle\langle\psi_i|\Big)\\ =& \sum_i p_i\frac{d}{dt}|\psi_i\rangle\langle\psi_i| + \sum_i p_i|\psi_i\rangle\frac{d}{dt}\langle\psi_i|\\ =& \sum_i p_i\Big(\frac{H}{i\hbar}|\psi_i\rangle\Big)\langle\psi_i| + \sum_i p_i|\psi_i\rangle\Big(\frac{H}{-i\hbar}\langle\psi_i\Big)\\ =& \frac{H}{i\hbar}\sum_i p_i|\psi_i\rangle\langle\psi_i| - \sum p_i|\psi_i\rangle\langle\psi_i|\frac{H}{i\hbar}\\ =& \frac{1}{i\hbar}[H,\rho] = -\frac{i}{\hbar}[H,\rho]\end{aligned}$$

We see that mixed states and pure states evolve under the same Hamiltonian law.

Remark. (Necessary condition for a pure state) Notice that the square of the density matrix is equal to

$$\rho^2 = \sum_j p_j^2|\psi_j\rangle\langle\psi_j|$$

Therefore, in general, we say mixed states are states whose density matrix satisfies

$$\hat{\rho} \neq \hat{\rho}^2$$

and a pure state otherwise.

Proof. Let's compare ρ^2 and ρ

$$\rho - \rho^2 = \sum_j (p_j - p_j^2)|\psi_j\rangle\langle\psi_j|$$

We see that the expectation value is always bigger than 0 (sometimes, we write $\rho \geq \rho^2$). We may also ask, what is the norm of $\rho|\psi\rangle$ for an arbitrary $|\psi\rangle$? We notice that

$$\langle\rho\psi|\rho\psi\rangle = \langle\psi|\rho^2|\psi\rangle = \langle\psi|\rho\psi\rangle \leq \sqrt{\langle\psi|\psi\rangle}\sqrt{\langle\rho\psi|\rho\psi\rangle}$$

Resulting in

$$\langle \rho\psi|\rho\psi\rangle \leq \langle\psi|\psi\rangle$$

We see that the density matrix is a contraction mapping. We now prove the contraposition, that if $\rho^2 = \rho$, the state is pure. Last class we have seen that the eigenvalue of the matrix becomes either 1 or 0. In the spectral decomposition,

$$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$$

we see that only one of the p_j's should become one and that the others should be zero. We have now concluded the proof. $\square$

Theorem. (Equivalence of representations) Consider an orthonormal basis $\{|\psi_i\rangle\}_{i=1}^n$ and a not necessarily orthonormal basis $\{|\phi_j\rangle\}_{j=1}^m$.

$$\rho = \sum_{i=1}^n p_i|\psi_i\rangle\langle\psi_i| \qquad \& \qquad \rho = \sum_{j=1}^m q_j|\phi_j\rangle\langle\phi_j|$$

As n is the minimum number of vectors the span the set, $m \geq n$. Define a $m \times n$ matrix

$$V_{ij} = \sqrt{\frac{q_i}{p_j}}\langle\psi_j|\phi_i\rangle$$

and we see that

$$\sum_{j=1}^m (V^\dagger)_{ij}V_{jk} = \sum_{j=1}^m \sqrt{\frac{q_j}{p_i}}\langle\phi_j|\psi_i\rangle\sqrt{\frac{q_j}{p_k}}\langle\psi_k|\phi_j\rangle = \sum_{i=1}^n \sqrt{\frac{p_i^2}{p_i p_k}}\langle\psi_i|\psi_i\rangle\langle\psi_k|\psi_i\rangle = \delta_{ik}$$

and therefore $V^\dagger V = I_n$. After adding $m-n$ column vectors to V, we can make it unitary, and it would arrive at

$$\sum_{j=1}^n U_{ij}\sqrt{p_j}|\psi_j\rangle = \sqrt{q_i}|\phi_i\rangle$$

This can be used to prove that ρ can indeed be expressed as $\sum_{i=1}^m q_i|\phi_i\rangle\langle\phi_i|$.

Lecture 3 (September 8th)

Definition. (Non-degenerate pertubation) The problem of non-degenerate pertbuation esstentially boils down matching orders of Taylor expansions. Assuming

$$E_n^{(0)}|n^{(0)}\rangle = H^{(0)}|n^{(0)}\rangle$$

we have

$$E_n(\lambda)|n\rangle_\lambda = H(\lambda)|n\rangle_\lambda$$

Taylor expanding,

$$(E_n^{(0)} + \lambda E_n^{(1)} + \ldots)|n\rangle_\lambda = (H^{(0)} + \lambda H^{(1)})c_n(\lambda)(|n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \ldots)$$

Matching the powers of one, we find

$$E_n^{(1)}|n^{(0)}\rangle + E_n^{(0)}|n^{(1)}\rangle = H^{(1)}|n^{(0)}\rangle + H^{(0)}|n^{(1)}\rangle$$

and using the projection operator,

$$|n^{(1)}\rangle = \frac{1}{E_n^{(0)} - H^{(0)}}\phi_n(H^{(1)}|n^{(0)}\rangle)$$

Matching the powers of two, we find

$$|n^{(2)}\rangle = \frac{1}{E_n^{(0)} - H^{(0)}}\phi_n(H^{(1)}|n^{(1)}\rangle) + \frac{1}{E_n^{(0)} - H^{(0)}}\phi_n(-E^{(1)}|n^{(1)}\rangle)$$

We are thus able to identify the higher orders of $|n\rangle_\lambda$ up to the second order, with

$$|n\rangle_\lambda = c_n(\lambda)\Big(|n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \lambda^2|n^{(2)}\rangle + \ldots\Big)$$

Lecture 4 (September 10th)

Definition. (Non-degenerate pertubation energies) We would now want to identify the higher orders of the energy, no the eigenvectors. We can applying $\langle n^{(0)}|$ to the expansion and repeating the process above, we obtain, for first order,

$$\begin{aligned} E_n^{(1)} =& \langle n^{(0)}|H^{(0)}|n^{(1)}\rangle + \langle n^{(0)}|H^{(1)}|n^{(0)}\rangle \\ =& \langle n^{(0)}|H^{(1)}|n^{(0)}\rangle \end{aligned}$$

and for the second order,

$$\begin{aligned}
E_n^{(2)} =& \langle n^{(0)}|H^{(0)}|n^{(2)}\rangle + \langle n^{(0)}|H^{(1)}|n^{(1)}\rangle \\
=& \langle n^{(0)}|H^{(1)}|n^{(1)}\rangle \\
=& \langle n^{(0)}|H^{(1)}\frac{1}{E^{(0)}-H^{(0)}}\phi_n(H^{(1)}|n^{(0)}\rangle) \\
=& \sum_{k\neq n}\langle n^{(0)}|H^{(1)}\frac{1}{E^{(0)}-H^{(0)}}|k^{(0)}\rangle\langle k^{(0)}|H^{(1)}|n^{(0)}\rangle \\
=& \sum_{k\neq n}\langle n^{(0)}|H^{(1)}|k^{(0)}\rangle\frac{1}{E^{(0)}-E_k^{(0)}}\langle k^{(0)}|H^{(1)}|n^{(0)}\rangle \\
=& \sum_{k\neq n}\frac{|\langle n^{(0)}|H^{(1)}|k^{(0)}\rangle|^2}{E_n^{(0)}-E_k^{(0)}}
\end{aligned}$$

Notice how we have considered $\langle n^{(0)}|H^{(0)}|n^{(i)}\rangle = 0$. This comes from the previous fact that higher orders came with a projection operator that elimiates the $|n^{(0)}\rangle$ in them.

I think we've done enough work worthy of memorising! The first and second Taylor coefficients for energy are essentially the expectation value of the perturbing potential and the weighted sum of coalescing states.

Example. (Simple harmonic oscillator) Consider the solved harmonic oscillator with basis vectors $|n^{(0)}\rangle$ and $E_n^{(0)} = \hbar\omega(n+1/2)$. Applying a uniform electric field, Ex,

$$\lambda H^{(1)} = (-e)(-Ex) = eEx$$

For the first order, we have

$$\lambda E_n^{(1)} = \langle n^{(0)}|\lambda H^{(1)}|n^{(0)}\rangle = \langle n^{(0)}|eE\sqrt{\frac{m\omega}{2\hbar}}(a+a^\dagger)|n^{(0)}\rangle = 0$$

For the second order, we have

$$\lambda^2 E_n^{(2)} = \sum_{k\neq n}\frac{1}{E_n^{(0)}-E_k^{(0)}}|\langle n^{(0)}|eE\sqrt{\frac{m\omega}{2\hbar}}(a+a^\dagger)|k^{(0)}\rangle|^2 = -\frac{(eE)^2}{2m\omega^2} \ < \ 0$$

We know that this result, however, is exact. This is because of the form of the Hamilonian,

$$H^{(0)} + \lambda H^{(1)} = \frac{p^2}{2m} + \frac{1}{2}m\omega^2x^2 + eEx = \frac{p^2}{2m} + \frac{m\omega^2}{2}(x-K) - \frac{(eE)^2}{2m\omega^2}$$

where K is a constant. Note that in the ground state,

$$\lambda^2 E_0^{(2)} = \sum_{k\neq n}\frac{|\langle n^{(0)}|\lambda H^{(1)}|k^{(0)}\rangle|^2}{E_n^{(0)}-E_k^{(0)}} \ < \ 0$$

Lecture 5 (September 15th)

Definition. (Degenerate pertubation theory) In the prior case, we've seen energy levels being strictly distinct. What if there are multiple states per energy level?

$$P_0 = \sum_D |l_0\rangle\langle l_0|$$

where D is a degenerate subspace. The identity can thus be written as

$$\mathbf{1} = P_0 + P_1$$

Obviously,

$$\hat{H}_0 P_0 = \hat{H}_0 \sum_l |l_0\rangle\langle l_0| = E_D^{(0)} P_0$$

and

$$P_0 \hat{H}_0 = \sum_l |l_0\rangle\langle l_0|\hat{H}_0 = E_D^{(0)} P_0$$

and the two operators commute. We ultimately want to solve

$$\hat{H}|l\rangle = (\hat{H}_0 + \lambda\hat{V})|l\rangle = E|l\rangle$$

we state that it can be expressed as

$$|l\rangle = P_0|l\rangle + P_1|l\rangle$$

Applying the Hamiltonian, we expect

$$(\hat{H}_0 + \lambda\hat{V})(P_0|l\rangle + P_1|l\rangle) = E(P_0|l\rangle) + E(P_1|l\rangle)$$

Applying once more the projection operator P_0,

$$(E_D^{(0)} + P_0\lambda\hat{V}P_0)P_0|l\rangle + P_0\lambda\hat{V}P_1|l\rangle = EP_0|l\rangle$$

we can of course subtract the most-left term and arrive at

$$(P_0\lambda\hat{V}P_0)P_0|l\rangle + P_0\lambda\hat{V}P_1|l\rangle = (E - E_D^{(0)})P_D|l\rangle$$

where the energy difference on the right is called the energy spitting δE_0. Ignoring the second term as we want only the 1st order, we have

$$(P_0\lambda V P_0)P_0|l\rangle = \delta E_0 P_0|l\rangle$$

We can also apply P_1 from the left to prove that indeed the second term is of second order and beyond. Having

$$\begin{aligned} P_1\lambda\hat{V}P_0|l\rangle + (\hat{H}_0 + P_1\lambda\hat{V}P_1)P_1|l\rangle =& EP_1|l\rangle \\ (E - \hat{H}_0 - P_1\lambda V P_1)P_1|l\rangle =& P_1\lambda V P_0|l\rangle \end{aligned}$$

which leads to

$$P_1|l\rangle = \frac{1}{E - H_0}P_1\lambda V P_0|l\rangle$$

Theorem. The following

$$\boxed{(P_0\lambda V P_0)P_0|l\rangle = \delta E_0 P_0|l\rangle}$$

is ultimately the eigenvalue problem we want to solve to find the perturbed eigenkets. The energy is resultantly

$$\boxed{E = E_0^{(0)} + \delta E_0 + \lambda\sum_{k\notin D}\frac{|V_{kl}|^2}{E_D^{(0)} - E_k^{(0)}}}$$

Example. (Isotropic rigid rotator) Consider the Hamiltonian

$$\hat{H}_0 = \frac{\vec{L}\cdot\vec{L}}{2I}$$

We know that the eigenkets and their energy levels are given as

$$|l, m\rangle \qquad \& \qquad \frac{l(l+1)\hbar^2}{2I}$$

If $l \neq 0$, degenerate $2l + 1$. The pertubation will be given by

$$\hat{V} = -\vec{\mu}\cdot\vec{B} = -\frac{gqB}{2m}L_z$$

where $\vec{B}$ is an external magnetic field and $\vec{\mu} = gq\vec{L}/2m$. The g is dicated by whether its electron spin (=2) or whether its orbitals (=1). These kind of potentials are from Zeeman coupling. Now, we see that the potential is also commutative with L_z, as $[V, L_z] = 0$. We should utilise this! For a general state vector,

$$\langle m|[V, L_z]|m'\rangle = \langle m|VL_z - L_zV|m'\rangle = 0$$

which implies that

$$(m' - m)\hbar\langle m|V|m'\rangle = 0$$

Thus we have found the matrix element for the pertubation! We thus find

$$\frac{l(l+1)\hbar^2}{2I} - x\hbar m$$

for $x > 0$.

Definition. (Stark effect of the hydrogen atom) Assume no spin. For an Hydrogen atom in an electric field in the z-axis,

$$H_1 = (-e)V(r) = (-e)(-Ez) = eEz$$

Lecture 6 (September 17th)

Chapter 12. The Real Hydrogen Atom

Lemma. (Relativistic kinetic energy) Recall that the relativistic energy kinetic energy was given as

$$E = \sqrt{(\vec{p}c)^2 + (mc^2)^2} - mc^2$$

Taylor expanding, we find

$$E = \frac{\vec{p}^2}{2m_e} - \frac{1}{8}\frac{(\vec{p}^2)^2}{m_e^3 c^2} + \dots$$

and we will use

$$H_1 = -\frac{1}{2}\left(\frac{\vec{p}^2}{2m_e}\right)^2 \frac{1}{m_e c^2}$$

Finding the first order pertubation for this potential, we find

$$\langle \phi_{nlm} | H_1 | \phi_{nlm} \rangle = -\frac{1}{2} m_e c^2 (Z\alpha)^2 \left[\frac{2(Z\alpha)^2}{n^3(2l+1)} - \frac{3(Z\alpha)^2}{4n^4}\right]$$

Lemma. (Spin-orbit coupling) The spin of the electron and its associated magnetic moment

$$\vec{\mu} = -\frac{e}{2m_e} g\vec{S} = -\frac{e}{m_e}\vec{S}$$

change the interaction of the electron with the electric field that is produced by the proton charge. An electron moving with a uniform velocity $\vec{v}$ experiences a magnetic field

$$\vec{B} = -\frac{\vec{v} \times \vec{E}}{c^2} = -\frac{\vec{v} \times \vec{E}}{c^2} = -\frac{\vec{v} \times (-\nabla\phi(r))}{c^2} = \frac{1}{c^2}\vec{v} \times \frac{\vec{r}}{r}\frac{d\phi(r)}{dr}$$

from here, the additional interaction becomes

$$H_2 = -\vec{\mu} \cdot \vec{B} = -\frac{e}{m_e}\vec{S} \cdot \vec{v} \times \frac{\vec{r}}{rc^2}\frac{d\phi(r)}{dr} = -\frac{e}{m_e^2 c^2}\vec{S} \cdot \vec{L}\,\frac{1}{r}\frac{d\phi(r)}{dr}$$

Taking $\phi(r) = Ze/(4\pi\varepsilon_0 r)$, we find

$$H_2 = \frac{Ze^2}{4\pi\varepsilon_0}\frac{1}{m_e^2 c^2}\frac{\vec{S} \cdot \vec{L}}{r^3}$$

where a relativistic calculation adds another factor of 1/2. We conclude

$$H_2 = \frac{Ze^2}{4\pi\varepsilon_0}\frac{1}{2m_e^2c^2}\frac{\vec{S}\cdot\vec{L}}{r^3}$$

Lecture 7 (September 23rd)

Definition. (Spin-orbit interaction) We call the splitting of energy levels due to the spin-orbit interactions the fine structure. The magnetic dipole energy due to the magnetic field is

$$-\vec{\mu}_e\cdot\vec{B}' = -\Big(\frac{-eg}{2m_e}\Big)\vec{S}\cdot\vec{B}'$$

by subtituting the magnetic potential, we find

$$\begin{aligned}-\vec{\mu}_e\cdot\vec{B}' =& \frac{eg}{2m_e}\vec{S}\cdot\Big(\frac{\vec{v}}{c^2}\times\frac{\vec{r}}{r}\frac{d\phi}{dr}\Big)\\ =& -\frac{eg}{2m_e^2c^2}\vec{S}\cdot\vec{L}\Big(\frac{1}{r}\frac{d\phi}{dr}\Big)\end{aligned}$$

where $\vec{L} = \vec{r}\times(m_e\vec{v})$. When done properly, we find Thomas precession, giving

$$\phi(\vec{r}) = \Big(\frac{ze}{r}\frac{1}{4\pi\varepsilon_0}\Big)$$

This ultimately gives

$$H_{SO} = \frac{1}{2}\Big(\frac{g}{2}\Big)\frac{ze^2}{4\pi\varepsilon_0}\frac{1}{2m_e^2c^2}\frac{\vec{S}\cdot\vec{L}}{r^3}$$

In the Hydrogen atom, now consider the eigenvectors $|n,l\rangle$, in addition to the spin as $|n,l,m,6\rangle$. We find $2(2l+1)$ degenerate levels. Recall that we simply need to diagonalise the matrix for degenerate levels

$$\langle l,m',6'|H_{SO}|l,m,6\rangle$$

Also recall that

$$H_{SO}\propto\vec{S}\cdot\vec{L} = \frac{1}{2}\Big(\vec{J}^2-\vec{L}^2-\vec{S}^2\Big)$$

and therefore

$$|l,m,5\rangle\to|l,j,m_j\rangle$$

should be the basis we should consider and it would be automatically diagonalised. Recall the following formula

$$|j=l+\frac{1}{2},m+\frac{1}{2}\rangle = \sqrt{\frac{l+m+1}{2l+1}}|l,m\uparrow\rangle+\sqrt{\frac{l-m}{2l+1}}|l,m+1\downarrow\rangle$$

Lecture 9 (September 22nd)

Proposition. (Two identical particles) Observe that the center of mass momentum and variables are

$$\vec{P} = \frac{m_2\vec{p}_1 + m_1\vec{p}_2}{m_1+m_2} = \frac{1}{2}(\vec{p}_1 - \vec{p}_2) \qquad \& \qquad \vec{r} = \vec{r}_1 - \vec{r}_2 \qquad \& \qquad \mu = \frac{m_1 m_2}{m_1+m_2}$$

For identical particles, the reduced mass becomes $\mu = m/2$, and in the center of mass coordinates, the Hamiltonian becomes

$$\left(E - \frac{\vec{P}^2_{\text{tot}}}{2M}\right)\psi(\vec{r}) = \left(\frac{\vec{p}^2}{2\mu} + V(|\vec{r}|)\right)\psi(\vec{r})$$

For two identical fermions, we observe the following constraints for $\psi(\mathbf{r})$:

(i) Under interchange,

$$\psi(\vec{r}) \mapsto -\psi(\vec{r})$$

(ii) And excluding spin and only the spatial part,

$$\psi \sim R_{nl}(r) Y_{lm}(\theta, \phi)$$

with Y_{lm} having $(-1)^l$ parity.

(iii)

$$\frac{1}{2} \otimes \frac{1}{2} = \text{ triplet symmetry } + \text{ singlet symmetry}$$

Therefore, since the total wavefunction must be antisymmetric, spin singlet (antimetric) $\times$ spatial symmetric is allowed requiring even l $(0, 2, 4, \ldots)$. Also, spin triplet (symmetric) $\times$ spatial antisymmetric is allowed requiring odd l $(1, 3, 5, \ldots)$.

Remark. (Intrinsic parity of fundamental particles)

(i) Let us first define the intrinsic parities of the electron, proton, and neutron as $+1$.

(ii) Let us find the parity of the hydrogen atom at $l = 1$. We find $(+1)(+1)(-1) = (-1)$.

(iii) The parity of particles and anti-particles are opposite. Notice that for the positronium, $(+1)(-1)(+1) = 0$.

(iv) As we know well the properties of the proton, neutron, and electron, we wanted to guess the properties of particles such as the pions (π^-, π^+, and π^0). Also, we wanted to know particles like the deutron ($d^+ = p^+ - n$) and deuterium ($D = d^+ - e^-$). After multiple particle experiments, we've found the capture reaction channel

$$d^+ + \pi^- \to n + n$$

and we have found that the total angular momentum of the initial system was 1 with the deuterium particle having spin 1. With the result being two identical fermions, we use the results above to analyse the final system. We have two possibilities, a spin singlet ($S = 0$) or spin triplet ($S = 1$). From spin conservation, we are able to know the specific state that the final state should be in.

The critical point of the last remark is that parity is conserved. In the last state, the parity is -1, and the inital state, we had 1 times the parity of π^{-1} telling us that the parity of π^{-1} is -1.

Theorem. (Pauli exclusion principle of non-interacting fermions) We look at how the Pauli exclusion principle works for electrons. Consider a electron confined in a box of length L. The wavefunction of a single particle and its energy can be written as (through multiplying all components for the solution of an infinite potential well)

$$\left(\frac{2}{L}\right)^{3/2} \sin\left(\frac{n_1 \pi x}{L}\right) \sin\left(\frac{n_2 \pi y}{L}\right) \sin\left(\frac{n_3 \pi z}{L}\right)$$

where $n_1, n_2, n_3 \geq 1$. The energy should be given as

$$\frac{\hbar^2}{2m}\left[\left(\frac{\pi n_1}{L}\right)^2 + \left(\frac{\pi n_2}{L}\right)^2 + \left(\frac{\pi n_3}{L}\right)^2\right]$$

and is independent of spin. Assume that there are 24 electrons inside the box. We want to see that the minimum energy is for these particles. In the lowest energy level of 3, there are 2 possible particles and for the second lowest energy level of 6, there are 6. Also,

$$E(N = 24) = 214 \times \left(\frac{\hbar^2 \pi^2}{2mL^2}\right)$$

for fermions with half-spin. What about for bosons? We would have all particles stuffed in the lowest state and would have

$$E(N = 24) = 72 \times \left(\frac{\hbar^2 \pi^2}{2mL^2}\right)$$

What is the number of states with an energy level lower than E? It is

$$\frac{1}{8} \frac{\text{Vol}}{\Delta n_x \Delta n_y \Delta n_z}$$

Lecture 10 (September 24th)

Recall. We have considered a box of side length L with energy levels denoted as

$$E_k = \frac{\hbar^2 \pi^2}{2mL^2}(n_1^2 + n_2^2 + n_3^2)$$

considering the number of states with energy level $E \leq E_k$. In a lattice structure, take $\Delta\vec{n} = 1$ with spin up and down. Taking

$$R^2 = n_1^2 + n_2^2 + n_3^2$$

we can take

$$R = \sqrt{\frac{E_k 2mL^2}{\hbar^2\pi^2}}$$

and express the number of energy levels inside a sphere with radius R lower than E_k as

$$2 \times \frac{1}{8} \times \frac{\frac{4\pi}{3}R^3}{\Delta\vec{n}} = N$$

Than what is the number of non-interacting electrons with the lowest energy state? We have

$$N_e = NE_F$$

where we call E_F is called the Fermi energy.

$$E_F = \frac{\hbar^2\pi^2}{2m}\Big(\frac{3n}{\pi}\Big)^{2/3} \qquad \& \qquad n = \frac{N_e}{L^3} = \frac{N_e}{V}$$

What is the lowest energy? It is

$$\frac{1}{8}\int_{E\leq E_F} d^3\vec{n}\, 2 \cdot \frac{\vec{p}^2}{2m} = \frac{1}{8}\int d^3\vec{n}\, 2\Big(\frac{\hbar^2\pi^2}{2m}\vec{n}^2\Big) = \frac{\hbar^2\pi^2}{10mL^2}R^5$$

substituting, we find

$$N_e\frac{3}{5}\frac{\hbar^2\pi^2}{2m}\Big(\frac{2n}{\pi}\Big)^{2/3} = \frac{3}{5}N_e F_E$$

Notice that we can write the Fermi energy in terms of a constant called the Fermi wave number,

$$E_F = \frac{\hbar^2 k_F^2}{2m}$$

Theorem. Taking this box system as a system of a quantum gas cloud, we can compute its pressure from its total energy as

$$P = -\Big(\frac{\partial E_{tot}}{\partial V}\Big)_{T=0}$$

A good way do this is by considering it's logarithm,

$$\ln E_{tot} = \ldots + \Big(-\frac{2}{3}\Big)\ln V$$

giving

$$\frac{1}{E_{tot}}\frac{\partial E_{tot}}{\partial V} = -\frac{2}{3}\frac{1}{V}$$

Notice how it doesn't follow the ideal gas law. We call this pressure ultimately as the degeneracy pressure.

Remark. In the atomic and molecular level, the number of considerations become emaculate:

(i) The kinetic energy of the electron

(ii) The attraction between the electron (e^-) and the nucleus (Ze)

(iii) The repulsion between the electrons

(iv) The spin-orbit interactions

Theorem. (Helium) We first consider the Helium atom with the Hamiltonian

$$H = \frac{\vec{p}_1^2}{2m_e} + \frac{\vec{p}_2^2}{2m_e} + \frac{1}{4\pi\varepsilon_0}\frac{(+Ze)(-e)}{r_1} + \frac{1}{4\pi\varepsilon_0}\frac{(+Ze)(-e)}{r_2} + \frac{(-e)(-e)}{4\pi\varepsilon_0}\frac{1}{r_{12}}$$

where the last term is $\hat{V}_{12}$ and the other terms are H_0 (Hyodrogen like). This is exactly soluble, and we need to Pauli exclusion principle for the two electrons. The Schrodinger equation that they will follow will be

$$H_0\psi(\vec{r}_1, \vec{r}_2, \sigma_1, \sigma_2) = E\psi(\vec{r}_1, \vec{r}_2, \sigma_1, \sigma_2)$$

and the energy would be the energy sum $E_1 + E_2$. The solutions will look something like

$$\phi_{n_1 l_1 m_1}(\vec{r}_1) \qquad \& \qquad \phi_{n_2 l_2 m_2}(\vec{r}_2)$$

The energy levels being

$$E_{n_1}^{(1)} = -\frac{Z^2}{n_1^2}(13.6) \qquad \& \qquad E_{n_2}^{(2)} = -\frac{Z^2}{n_2^2}(13.6)$$

depending on what energy level the electrons are in. In the lowest energy state for both particles, we simply add the lowest energies,

$$E_{grd} = E^{(1)} + E^{(2)} = 2E_1 = 2 \times (-54.5) = -108.8$$

where the units would be electron-volts. The possible spin states are the singlets ($S = 0$) or the triplets ($S = 1$). However, ote that as there are two particles, we cannot expect the wave functions to be odd under exchange, and the triplet state canont exist due to the Pauli exclusion principle. We automatically conclude that the spin state is a singlet!

We next consider the first excited state. The total energy should be the sum of the energy state of a $(1, 0, 0)$ state and a $(2, l, m)$ state giving

$$E_1 + E_2 = -54.4 - 13.6 = -68.0$$

We would have the wavefunctions, on the other hand, as

$$(1,0,0)(r_1)(2,l,m)(r_2) \pm (1,0,0)(r_2)(2,l,m)(r_1)$$

Definition. (Ionisation energy) The ionisation energy is the difference between energies as an electron escapes the molecule. Finding the difference of energies between two particles with energy -54.4 and just one particle at that level, we find $\Delta E_{ion} = +54.4$. Consider how when two particles were in the second energy state, $n = 2$ excited from the first, we find the difference of energy levels to be $\Delta E = -27.2 + 54.4 = 28.2$. This tells us that it requires less energy to ionize then to be excited to a new state! We call this autoionzisation.

Lecture 11 (September 29th)

Example. (Effects of electron-electron repulsion) In the presence of the a potential, the electron-electron Coulomb repulsion can be treated as a pertubation. The energy shift is given as

$$\Delta E = \iint d^3\vec{r}_1 d^3\vec{r}_2 \left(\phi_{100}(\vec{r}_1)\phi_{100}(\vec{r}_2)\right)^* \frac{e^2}{2\pi\varepsilon_0|\vec{r}_1 - \vec{r}_2} \left(\phi_{100}(\vec{r}_1)\phi_{100}(\vec{r}_2)\right)$$

Notice that this reduces to

$$\Delta E = \iint d^3\vec{r}_1 d^3\vec{r}_2 \, |\phi_{100}(\vec{r}_1)|^2 |\phi_{100}(\vec{r}_2)|^2 \frac{e^2}{2\pi\varepsilon_0|\vec{r}_1 - \vec{r}_2|}$$

This can be worked out to give

$$\Delta E = \frac{5}{8}\frac{Ze^2}{2\pi\varepsilon_0 a_0} = \frac{5}{4}Z\left(\frac{1}{2}mc^2\alpha^2\right)$$

as we know the zeroth order result to be -108.8 eV, we find up to first order, $E = -74.8$ eV.

Example. (First excited state) As we have two states with equal energies (spin triplet and spin singlet) we need to employ degenerate pertubation and diagonalise the 2×2 matrix in the degenerate subspace. The matrix will look like, for

$$\psi_\pm = \frac{1}{\sqrt{2}}\left(\phi_{100}(\vec{r}_1)\phi_{2lm}(\vec{r}_2) \pm \phi_{100}(\vec{r}_2)\phi_{2lm}(\vec{r}_1)\right) \cdot X_{s,t}$$

The matrix is expected to look like

$$\begin{pmatrix} \langle +|V_{12}|\rangle & \langle +|V_{12}|-\rangle \\ \langle -|V_{12}|+\rangle & \langle -|V_{12}|-\rangle \end{pmatrix}$$

However, the off-diagonal terms become zero and the matrix is already diagonal. We find the diagonal terms to be

$$\begin{aligned}\langle\pm|V_{12}|\pm\rangle =&\int d^3\vec{r}_1 d^3\vec{r}_2\,\psi^*_\pm(\vec{r}_1,\vec{r}_2)\frac{e^2}{4\pi\varepsilon_0|\vec{r}_1-\vec{r}_2|}\psi_\pm(\vec{r}_1,\vec{r}_2)\\ =&J_{nl}+K_{nl}\end{aligned}$$

where J_{nl} comes from the squares and K_{nl} comes from the cross terms.

$$\begin{cases}\Delta E^{(s)}_{nl} = J_{nl}+K_{nl}\\ \Delta E^{(t)_{nl}} = J_{nl}-K_{nl}\end{cases}$$

We find that the singlet state has a higher energy than the triplet state and that the spin had influence on the energy shift. We call this the spin exchange effect. Using Pauli matrices, we can write

$$\Delta E = J_{nl} - \frac{1}{2}(1+\vec{\sigma}_1\cdot\vec{\sigma}_2)K_{nl}$$

This develops to become Heisenberg's model of quantum magnetism.

Remark. For the ground state, we discover that $(L=0, S=0, J=L+S=0)$. Using the spectroscopy notation ${}^{2S+1}L_J$ we find ${}^1S_0$ and we call this state parahelium. For the first excited state, we note that $(100)(2s)$ is lower in energy than the higher $(100)(2p)$ due to the orbital shape. Each are denoted ${}^1S_0$ and ${}^3S_1$ where the latter is called ortho helium.

Lecture 12 (October 1st)

Definition. (Ritz variational principle) We are trying to find a good trial wavefunction $|\psi\rangle$ with some parameters for the ground state. Afterwards, we want to calculate $\langle\psi|\hat{H}|\psi\rangle$ and minimise this. Let

$$|\psi\rangle = \sum_{m=0} c_m|m\rangle$$

and $|m\rangle$ be $\hat{H}$'s true energy eigenkets which we do not know.

$$\langle\psi|\hat{H}|\psi\rangle = \sum_{n,m} c^*_n c_m\langle n|\hat{H}|m\rangle = \sum_m |c_m|^2 E_m \geq \sum_m |c_m|^2 E_0 \geq 0$$

where $\hat{H}|m\rangle = E_m|m\rangle$. We there for find that the expectation value is always higher than the true ground state and that the ground state acts as an lower bound!

Example. (Z^* hydrogenlike problem) We try to find the ground state using the trial ground state wavefunction as follows

$$\Big(\frac{\vec{p}^2}{2m_e}+\frac{(-e)(Z^*e)}{4\pi\varepsilon_0 r}\Big)\phi_{Z^*100}(r) =_g \phi_{Z^*100}(r) = \Big(-\frac{1}{2}m_e c^2\alpha^2\Big)(Z^*)^2$$

however, we expect

$$\psi_{Z^*}(\vec{r}_1, \vec{r}_2) = \phi_{Z^*100}(\vec{r}_1)\phi_{Z^*100}(\vec{r}_2)$$

From the hydrogen atom hamiltonian,

$$\hat{H} = \frac{\vec{p}_1^2}{2m_e} + \frac{\vec{p}_2^2}{2m_e} + \frac{Ze(-e)}{4\pi\varepsilon_0 r_1} + \frac{Ze(-e)}{4\pi\varepsilon_0 r_2} + \frac{e^2}{4\pi\varepsilon_0}\frac{1}{|\vec{r}_1 - \vec{r}_2}$$

we add and substract $Z^*(-e^2)/(4\pi\varepsilon_0 r_1)$ and we find

$$\langle\psi|\hat{H}|\psi\rangle = 2_g - 2\frac{(Z - Z^*)}{4\pi\varepsilon_0}\langle\frac{1}{r}\rangle + \frac{e^2}{4\pi\varepsilon_0}\langle\frac{1}{|\vec{r}_1 - \vec{r}_2|}\rangle$$

giving $Z^*_{\text{min}} = Z - 5/16$. Comparing our $E_{\text{min}} = -77.38$ eV and the experimental $E_{\text{exp}} = -78.975$ eV, we find a large accuracy.